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Roll No

EX-6002-CBGS

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Power System-II

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Why the interconnected power system is necessary?
Explain the problems associated with them.
b) What is deregulation in power system? Explain its effect
and how it can be overcome.
2. The following is the system data for a load flow solution. The
line admittances:

Bus code	Admittance
1 - 2	$2 - j8.0$
1 - 3	$1 - j4.0$
2 - 3	$0.666 - j2.664$
2 - 4	$1 - j4.0$
3 - 4	$2 - j8.0$

The schedule of Active and Reactive Power:

Bus code	P	Q	V	Remarks
1	-	-	1.06	Slack
2	0.5	0.2	$1 + j0.0$	PQ
3	0.4	0.3	$1 + j0.0$	PQ
4	0.3	0.1	$1 + j0.0$	PQ

Determine the voltages at the end of first iteration using
Gauss-Seidel Method. Take $\alpha = 1.6$.

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3. Develop load flow equations suitable for solution by Newton-Raphson method using nodal admittance approach.
4. Discuss in detail about the generation and absorption of reactive power in power system components.
5.
 - a) Derive an expression for the critical clearing angle for a power system consisting of a single machine supplying to an infinite bus, for a sudden load increment.
 - b) What are the various applications of equal area criterion?
6.
 - a) Derive the swing equation of a synchronous machine.
 - b) What is equal criteria? Explain.
7.
 - a) Explain with the help of block diagram automatic voltage regulator of turbo generators.
 - b) What is the need of the excitation system? Explain A.C. static excitation system.
8. Write short notes on any two:
 - a) Neutral grounding in the power system
 - b) Types of buses in load flows study
 - c) Power system stability
